Horizontal geothermal systems at Carleton College

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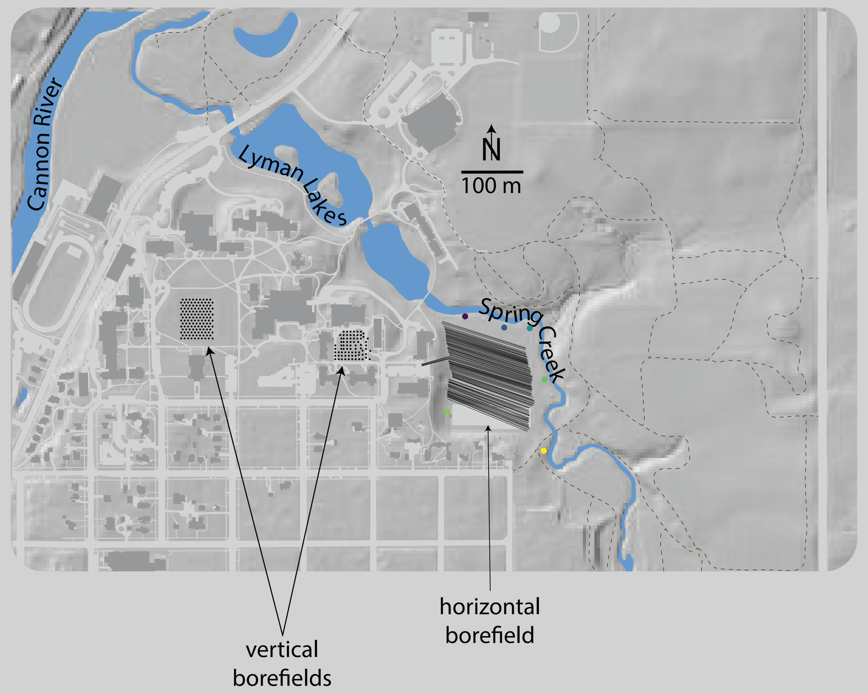
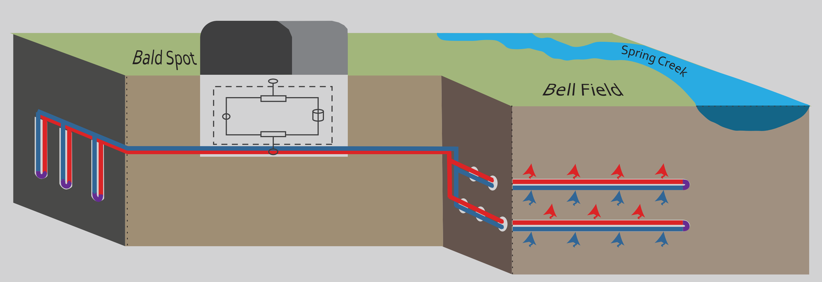
Carleton's geothermal system



In 2018, Carleton installed a district-scale closed-loop geothermal heat exchange system to heat and cool campus buildings.
It's powered by three borefields, one of which (Bell Field) consists of shallow, horizontal bores in unconsolioated sediment .

How do geothermal heat exchange systems work?



The ground should maintain a steady temperature around 10°C throughout the year. Carleton circulates fluid underground to maintain a constant supply of ambient temperature water, which heat exchangers split into hot and cold water. In the summer, the cold water is circulated to cool buildings, and in the winter, the opposite is true.

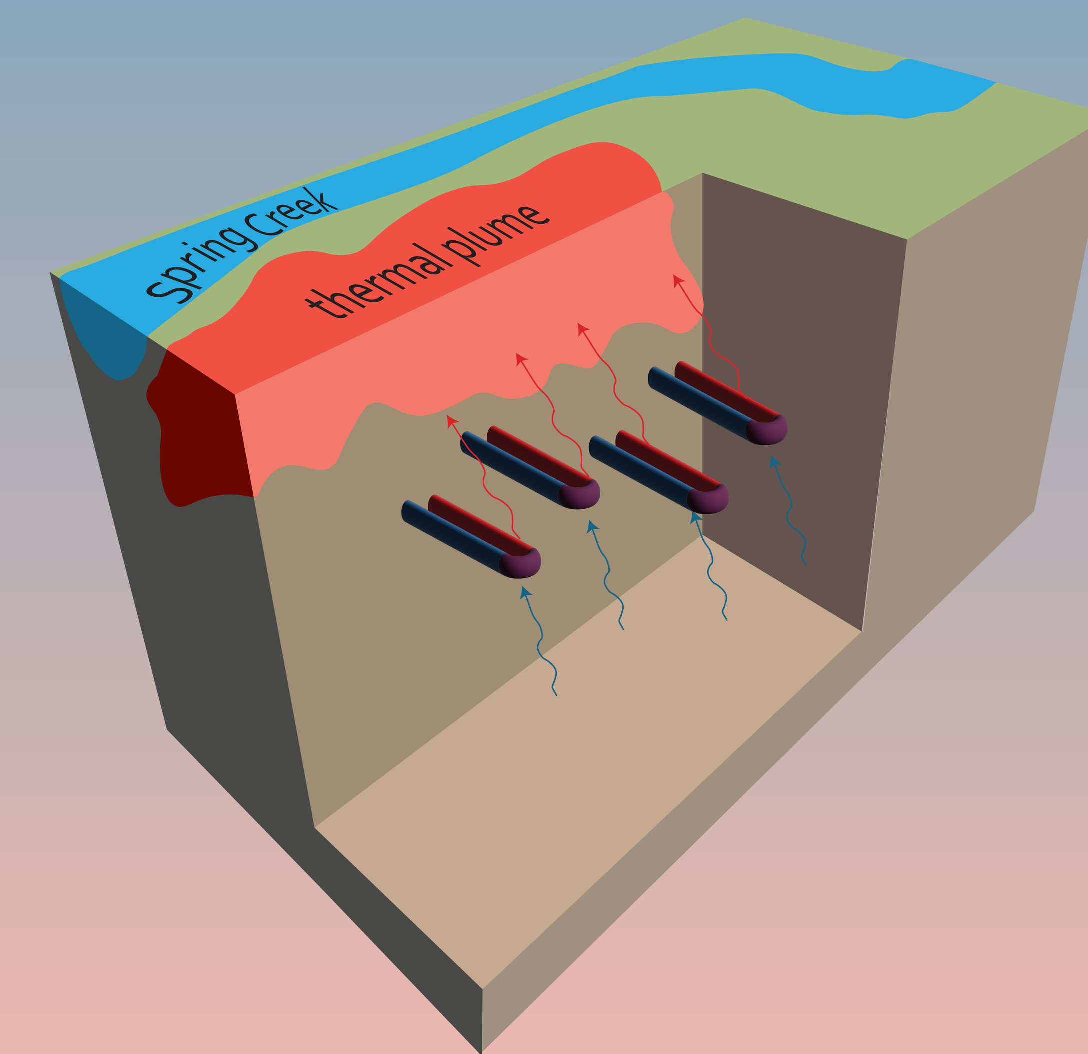
What is a thermal plume?

As heat is released into the borefield in the summer or cold water is circulated in the winter, we risk changing the temperature of the groundwater too quickly for it to recover.

Groundwater in the borefield might not move quickly enough for the geothermal system to maintain a steady temperature.

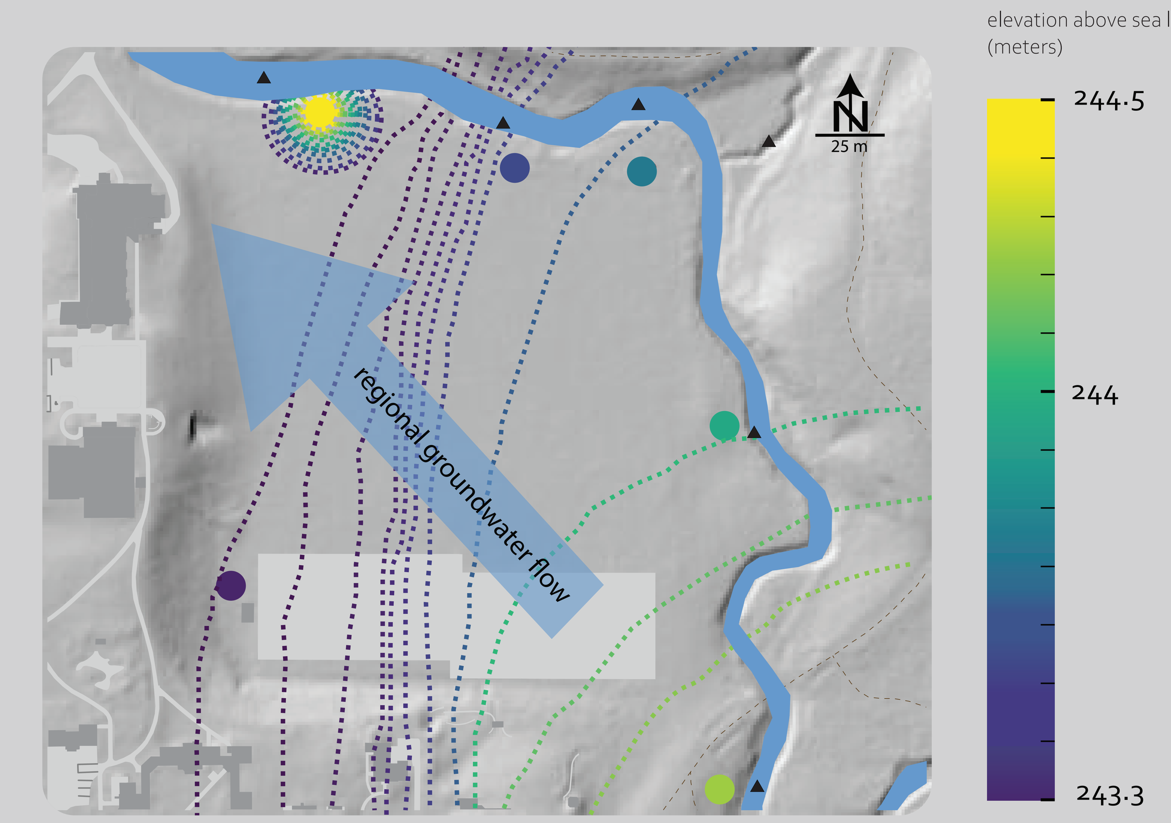
If a heat or cold plume develops, groundwater temperature will become more extreme over the season, risking the efficiency of the system.

A heat plume is expecially dangerous because it threatens dissolving toxic minerals into the aquifers, which provide drinking water for the town and school in addition to unstudied ecological effects.



The effects of a thermal plume will be most pronounced downgradient from the borefield.

Local groundwater flow



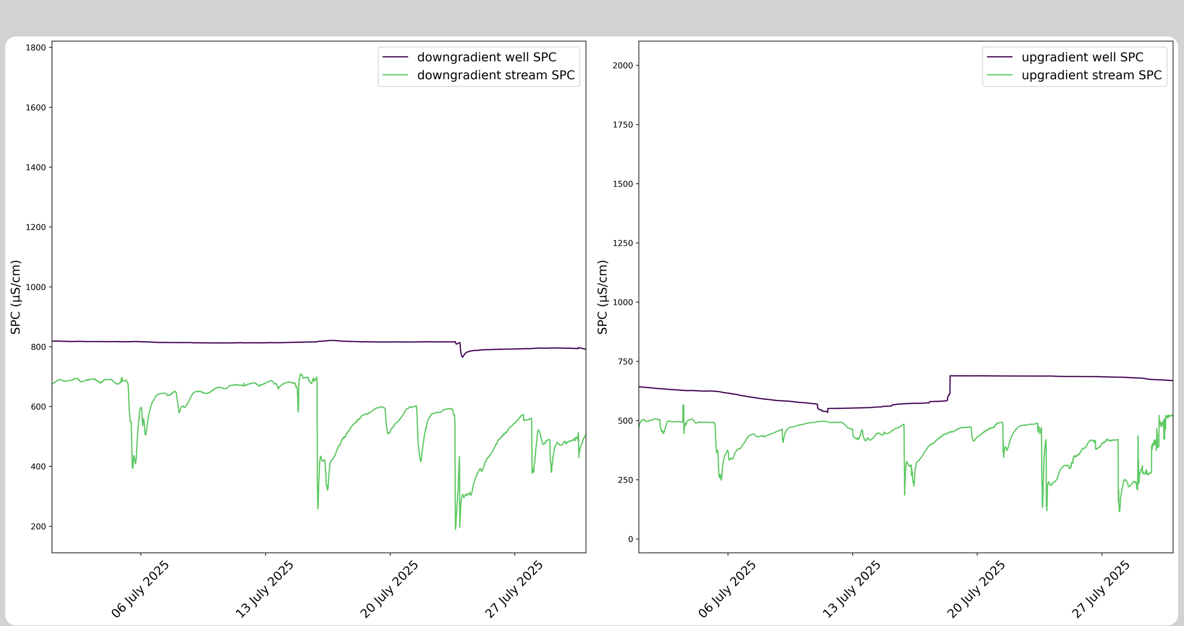
Regionally, grounwater moves northwest through campus to the Cannon River. However, the borefield is borderd on its north and east by Spring Creek, a small groundwater-fed stream.

Using water table elevation, water chemistry, and the known regional groundwater flow direction, we interpolate a groundwater surface and estimate a flow direction through the borefield.

Monitoring for temperature change

We installed monitoring wells along the egde of Bell Field, where we record temperature and dissolved solids to look for any thermal impacts of the geothermal system on the shallow groundwater.

How do we know the monitoring wells can measure the borefield?



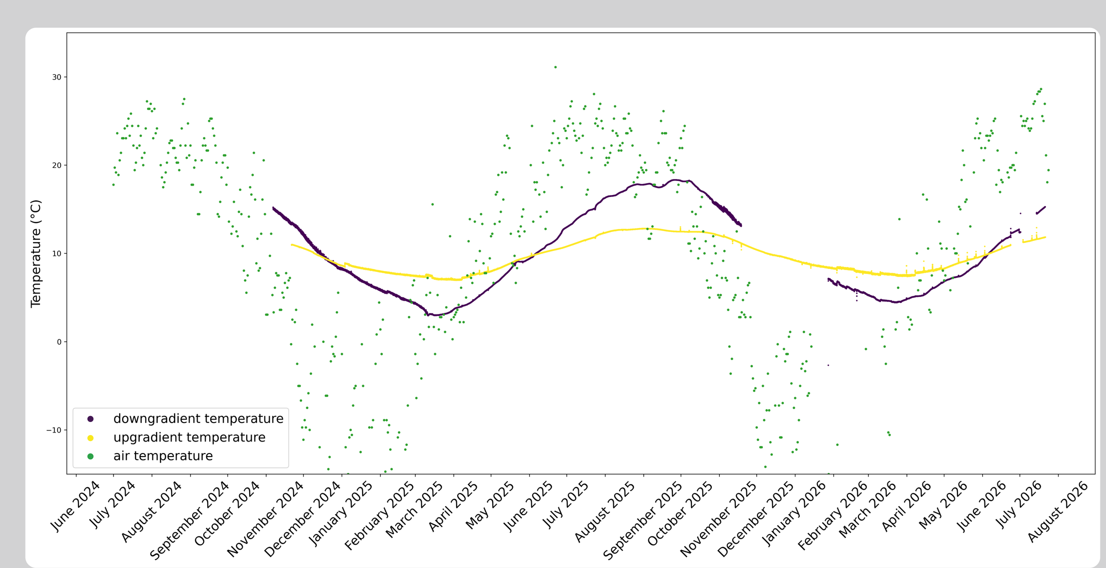
The dissolved solids in the creek vary seasonally, which we can trace to specific events, like precipitation, periods of road salting, or fertilizing the field.

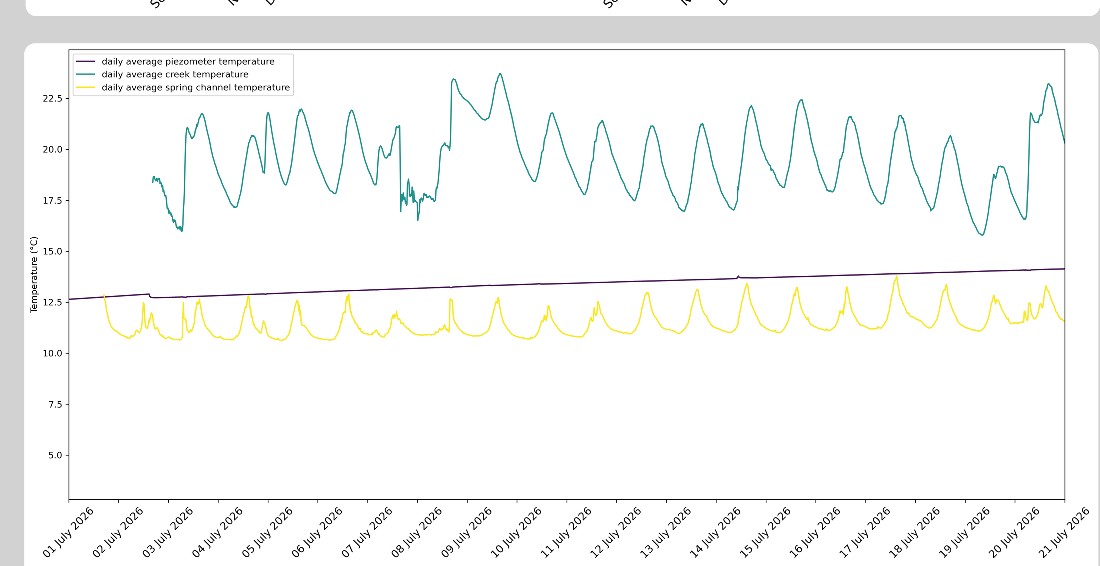
Well water varying with the creek would indicate that the wells reflect surface water or recieve an inflow of water from the stream.

Dissolved solids are significantly more stable in the wells than in the creek, indicating that the wells reflect deeper groundwater.

Evidence for thermal influence

Does the horizontal borefield create a thermal plume?



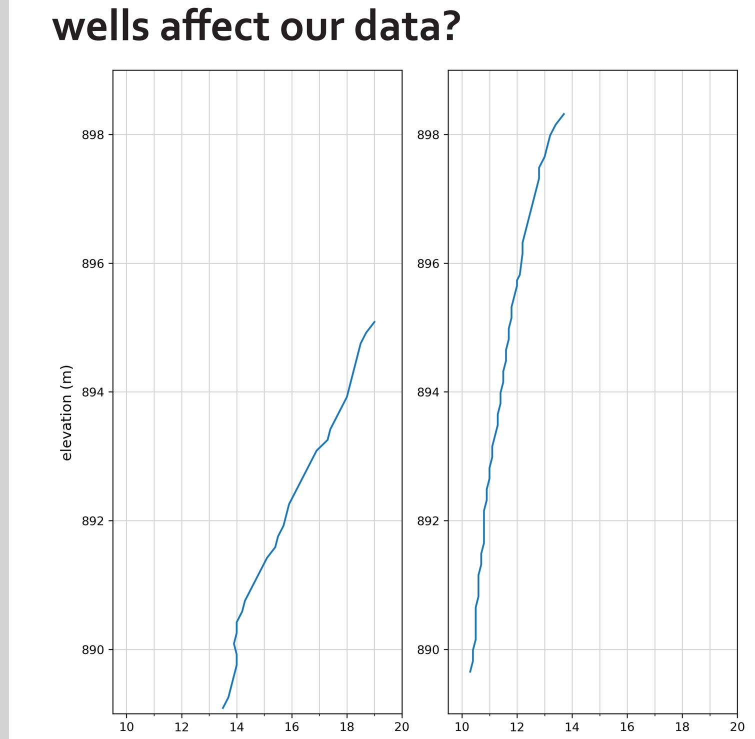


Temperature measurements across the monitoring wells reveal the possibility of the downgradient wells showing considerable impact from the geothermal system.
Especially during summertime, when the cooling load is heaviest, we speculate that the borefield warms the shallow groundwater multiple degrees.

The monitoring well farthest upgradient stays closer to 10° C throughout the year, and the farthest downgradient well significantly warms in the summer and cools in the winter.

We suspect that the stream water temperature is buffered by groundwater flow from the opposite bank.

Does the depth of the monitoring wells affect our data?



Yes, but the summer heat plume appears to shift downgradient temperature more than depth can explain.
In late July, the upgradient piezometer is colder at the same elevations.

Overall, we see evidence of a seasonal thermal plume but no evidence of it becoming more pronounced over time.